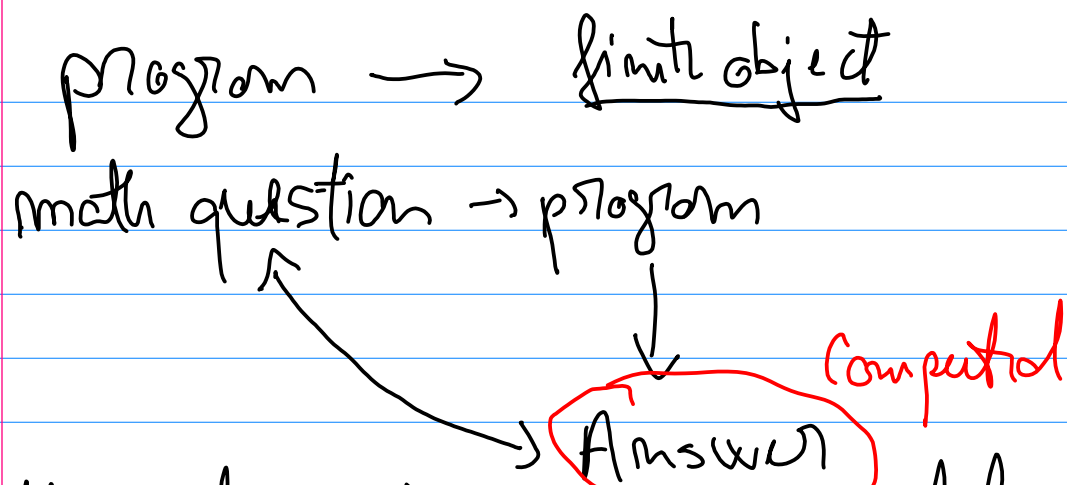




How does the answer relate to the Exact Answer

Approximation: proposes a value

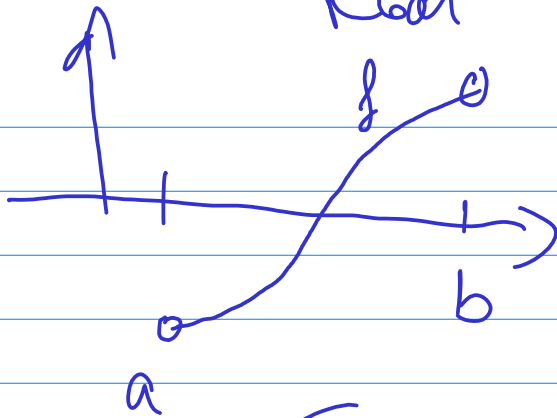
Error estimate: how accurate are we?

Enclosure: exact result belongs to an interval $[a, b]$

$\sqrt{2}$: $\sqrt{2} \in [1.4142, 1.4143]$

Certificate: claim, evidence (computation) mathematics.

Root $\exists x \in [a, b] \quad f(x) = 0$



- f continuous

Existence of root

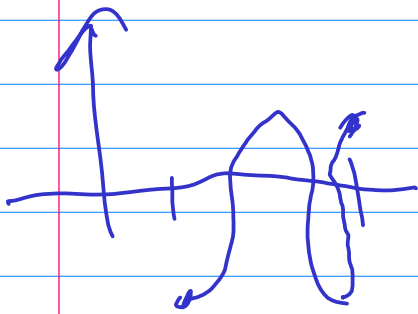


- f continuous (math)

- $f(a) \cdot f(b) \leq 0$

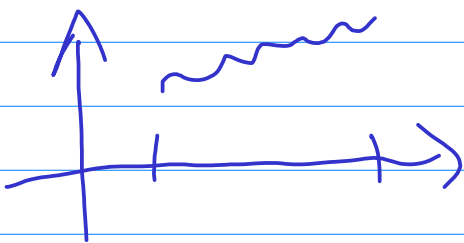
↑
computed.

Existence of unique root



I need more info.

No root here



- I need to check
all points in the
interval

Input representation

$0.1 \neq \frac{1}{10}$ in floating point

Rounding:

$a+b$
float



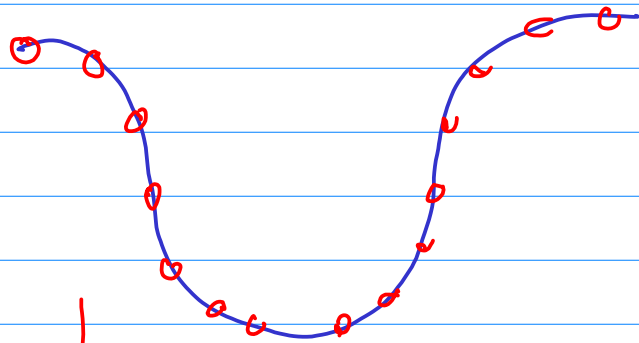
Floating point numbers

Truncation

$$10^{16} + 1 = 10^{16}$$

after the significant digits

Discretization



Looking at a finite number of points might lose "continuous" information.

Model discrepancy

vs
Newton gravity approximate at usual scales (speed)
Einstein gravity

Increasing computation precision $\neq$ a proof

Absolute errors: $|y - y_{\text{computed}}|$

- if y, y_{comp} are small $\rightarrow$ error is small

Relative errors $\frac{|y - y_{\text{comp}}|}{|y|}$ is not

necessarily small when y is small

Root certificate

Look for a root of $f(x) = x^2 - 2$

$x \in [a, b]$

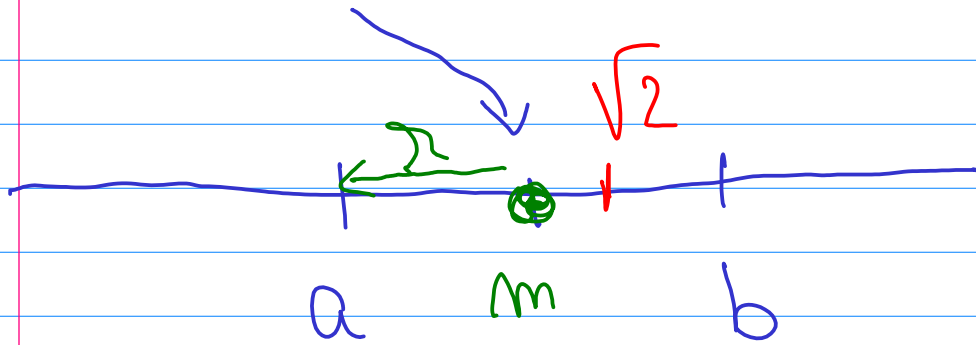
- math $\Rightarrow f$ continuous

- Compute $f(a) \Rightarrow f(a) < 0$ (SURE)

- Compute $f(b) \Rightarrow f(b) > 0$ (SURE)

- math $\Rightarrow f' = 2x > 0$, $f \nearrow$ on $[a, b]$

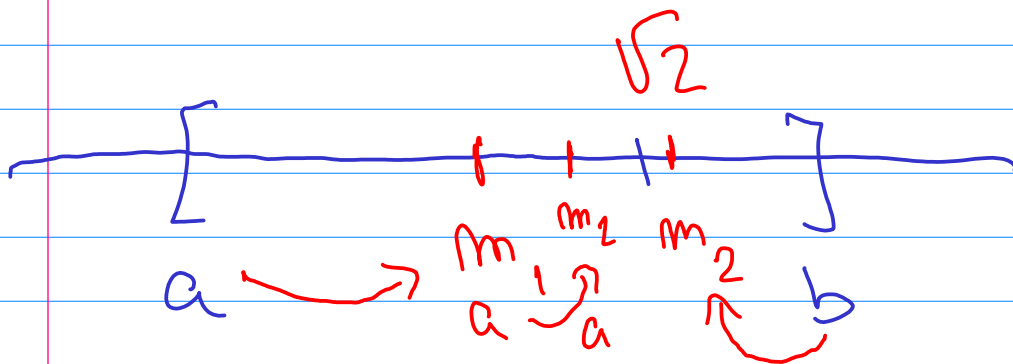
$$a^2 < 2 < b^2$$



What do I take as approx of $\sqrt{2}$?

$\sqrt{2} \approx m$ with error estimate ϵ

$$\sqrt{2} \in [m - \epsilon, m + \epsilon]$$



→ For approximating $\sqrt{2}$ we want

Residual $(x^2 - 2) \rightarrow$ small

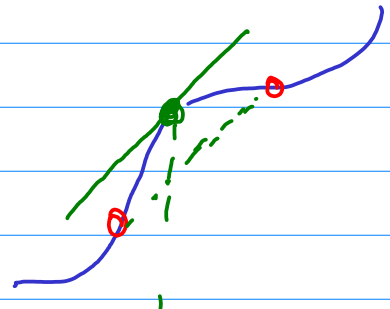
→ Attention: $h(x) = \underbrace{10^{-12}}_{\text{small}} (x - 1)$

$$|h(0)| = 10^{-12} \text{ small}$$

but 0 is far from the actual root which is 1.

Residual $\rightarrow$ Error

$\uparrow$
math



$$\underbrace{f(\text{approx}) - f(\sqrt{2})}_{=0} = \underbrace{f'(\xi)}_{\geq 2} \cdot (\text{approx} - \sqrt{2})$$

ξ

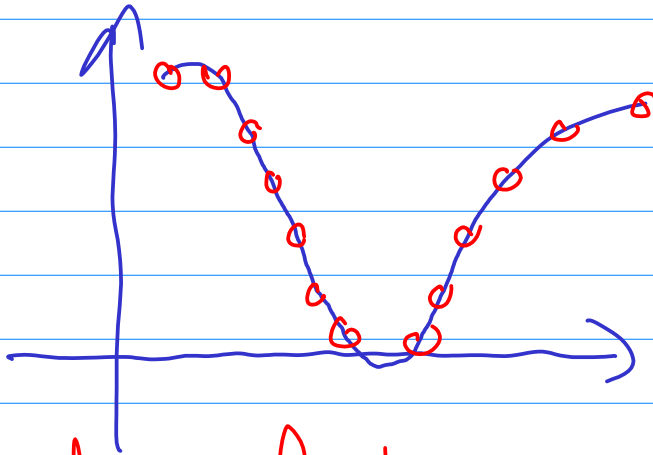
$$\Rightarrow \underbrace{|f(\text{approx})|}_2 \geq |\text{approx} - \sqrt{2}|$$

For a general problem we have the following errors:

- Continuous $\xrightarrow{\text{math}}$ Discrete (computations)
- Errors in the computer work
- Rounding errors ($0.1 + 0.2 \neq 0.3$ in float)

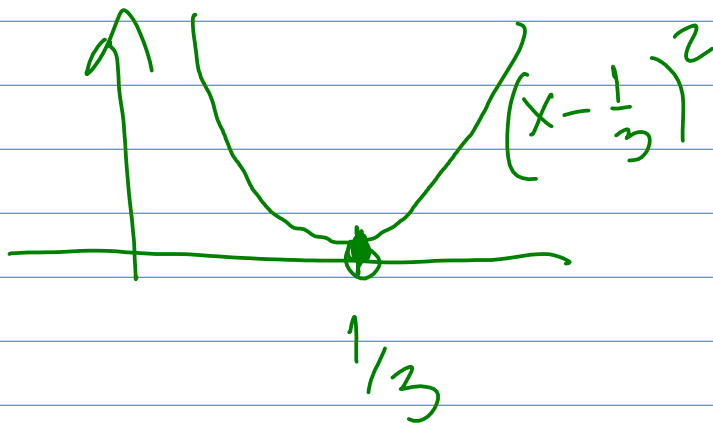
Certification: Control every source of error.

Grid results $\neq$ actual proof

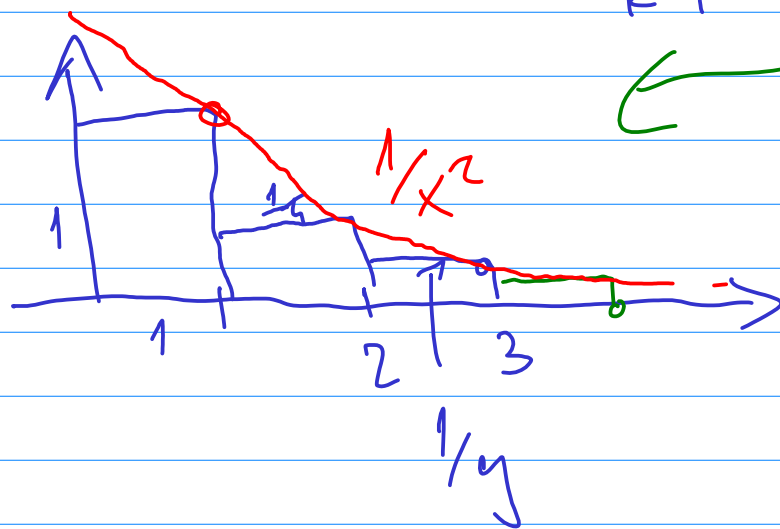


- function looks positive on a grid
- is not positive in a continuous setting.

A numerical observation $\neq$ a proof.



Section 5: Approx $\sum_{k=1}^{+\infty} \frac{1}{k^2} = S$



$$S \leq \int \frac{1}{x^2}$$

$$S_N = 1 + \dots + \frac{1}{N^2}$$

$$S_N + \frac{1}{N+1} \leq S \leq S_N + \frac{1}{N}$$

Explicit enclosure for S from S_N

Working with rationals is an easy solution, but not feasible for complex computations $\rightarrow$ large numerators or denominators.

Goal: advantages of floating pt comp.

- efficient memory
- speed

keep track of all roundoff errors
⇒ Interval arithmetic framework

$x \rightarrow [\underline{x}, \bar{x}]$
floating point numbers.